**DE–214** **13**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

CLASSICAL ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Examine the first five terms of the following series $\sum_{n=1}^{\infty} \frac{\cos nx}{n^2 + x^2}$. (10)

(b) (i) Prove : If $\sum c_n$ converges and if $\lim_{n\to\infty}\left(\frac{a_n}{c_n}\right)$ exists and is finite then $\sum a_n$ also converges. (5)

(ii) Prove : If $\sum d_n$ diverges and if $\lim_{n\to\infty}\left(\frac{a_n}{d_n}\right)$ exists and is greater than zero then $\sum a_n$ diverges. (5)

2. (a) Test the convergence of the hyper geometric series :

$$1 + \frac{\alpha\beta}{r}x + \frac{\alpha(\alpha+1)\beta(\beta+1)}{r(r+1)2!}x^2 + \ldots.$$ (10)

(b) State and prove Cauchy's root test in details. (10)

3. (a) (i) Examine the binomial series : (5)

$$1 + \frac{m}{1!}x + \frac{m(m-1)}{2!}x^2 + .... + \frac{m(m-1)...(m-n+1)}{n!}x^n + ...$$

(ii) Examine the logarithmic series : (5)

$$x - \frac{x^2}{2} + \frac{x^3}{3} - ... + (-1)^{n-1}\frac{x^n}{n} + ...$$

(b) Show that : (10)

$$\log\sqrt{2} = 1 + \left(\frac{1}{2} + \frac{1}{3}\right)\frac{1}{4} + \left(\frac{1}{4} + \frac{1}{5}\right)\frac{1}{4^2} + \left(\frac{1}{6} + \frac{1}{7}\right)\frac{1}{4^3} + ...$$

4. (a) Let $\sum a_n x^n$ be the given power series, let $\alpha = \lim\sup|a_n|^{1/n}$ and $R = \frac{1}{\alpha}$. (if $\alpha = 0$, $R = \infty$ and if $\alpha = \infty$, $R = 0$). Then $\sum a_n x^n$ converges absolutely if $|x| < R$, if $|x| > R$ the series is not convergent. (10)

(b) Sum the series to infinity

$$\frac{15}{16} + \frac{15.21}{16.24} + \frac{15.21.27}{16.24.32} + ...$$ (10)

5. (a) If $\alpha$ is an imaginary root of $x^7 - 1 = 0$, form the equation whose roots are $\alpha + \alpha^6$, $\alpha^2 + \alpha^5$, $\alpha^3 + \alpha^4$. (10)

(b) If $\alpha, \beta, \gamma$ are the roots of $x^3 + px^2 + qx + r = 0$, find the value of $\sum \alpha(\beta + \gamma)$. (10)

6. (a) Solve the equation

$$2x^5 - 15x^4 + 37x^3 - 37x^2 + 15x - 2 = 0.$$ (10)

(b) Solve the equation

$$x^6 - 4x^5 - 11x^4 + 40x^3 + 11x^2 - 4x - 1 = 0$$

having given that one root is $\sqrt{2} + \sqrt{3}$. (10)

7. (a) Prove that $1!\ 3!\ 5!\ ....\ (2n-1)! > (n!)n$. (10)

(b) State and prove Weierstrass' inequalities in details. (10)

8. (a) Solve the following simultaneous equations : (10)

$$x + 2y + 5z = 23$$
$$3x + y + 4z = 26$$
$$6x + y + 7z = 47.$$

(b) Diagonalise the matrix $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$. (10)

**DE–215** **14**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If $x = \sin(t)$ and $y = \sin(pt)$, then prove that $(1-x^2)y^2 - xy_1 + p^2 y = 0$. (10)

(b) If $V = r^m$, where $r^2 = x^2 + y^2 + z^2$, show that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = m(m+1)r^{m-2}$. (10)

2. (a) Show that the maximum value of $\left(\frac{1}{x}\right)^x$ is $e^{1/e}$. (10)

(b) Find the envelope of the family of straight lines, $\frac{x}{a} + \frac{y}{b} = 1$, where $a^2 + b^2 = k^2$. (10)

3. (a) (i) Evaluate : $\int \frac{dx}{x^2 + a^2}$. (5)

(ii) Evaluate : $\int \frac{dx}{\sqrt{a^2 - x^2}}$. (5)

(b) (i) Integrate the functions w.r.to $x$ : (5)

$$\frac{1}{a\sin x + b\cos x}.$$

(ii) Evaluate $\int \cos^{-1} x \; dx$ . (5)

4. (a) Evaluate $\int e^x \frac{1-\sin x}{1-\cos x} dx$ . (10)

(b) Obtain a reduction formula for $\int x^m (\log x)^n \; dx$ and apply it to evaluate $\int_0^1 x^4 (\log x)^3 \; dx$ . (10)

5. (a) Solve : $(1+y^2)\,dx + (x - \tan^{-1} y)\; dy = 0$ . (10)

(b) Solve : $\frac{dy}{dx} + y\cos x = 4x \operatorname{cosec} x$ , given that $y = 0$ when $x = \frac{1}{2}\pi$ . (10)

6. (a) Solve : $x^2 y'' - xy' + 4y = \cos(\log x) + x\sin(\log x)$ . (10)

(b) Solve : $(2x+1)^2\, y'' - 2(2x+1)\, y' - 12\, y = 6x$ . (10)

7. (a) Find the Laplace transform of : (10)

(i) $x\, e^{-x} \cos x$

(ii) $x^2 \cosh ax$

(iii) $x^2\, e^{-ax}$ .

(b) Using Laplace transforms, solve $xy''(2+x)\, y' + 3y = x - 1$ when $y(0) = 0$ . (10)

8. (a) Solve : $4(1+z^3) = 9z^4 pq$ . (10)

(b) Derive the working rule for 'Charpit's method'. (10)

**DE–216** **15**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

ANALYTICAL GEOMETRY AND VECTOR CALCULUS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) For what value of $k$ does $12x^2 + 7xy + ky^2 + 13x - y + 3 = 0$ represent a pair of straight lines? Find also the angle between them. (10)

   (b) Show that the circles $x^2 + y^2 - 4x + 6y + 8 = 0$ and $x^2 + y^2 - 10x - 6y + 14 = 0$, touch and find the co-ordinates of the point of contact. (10)

2. (a) If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two straight lines equidistant from the origin. Show that $f^4 - g^4 = c\left(bf^2 - ag^2\right)$. (10)

   (b) Trace the curve $\frac{10}{r} = 3\cos\theta + 4\sin\theta + 5$. (10)

3. (a) Prove : Length of the perpendicular from a point $A\left(x_1, y_1, z_1\right)$ to the plane $ax + by + cz + k = 0$ is $\pm\left[\frac{ax_1 + bx_1 + cz_1 + d}{\sqrt{(a^2 + b^2 + c^2)}}\right]$. (10)

   (b) Find the equation of the plane parallel to the plane $x + y + 2z = 1$ and through the point $\left(1, 1, -1\right)$. (10)

4. (a) Find the equation of the plane passing through the line of intersection of the planes $2x - y + 5z - 3 = 0$ and $4x + 2y - z + 7 = 0$ and parallel to the $z$-axis. (10)

(b) Find the image of the point $(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$. (10)

5. (a) Show that the surface generated by a straight line which meets the given skew lines at the same angles is $yz = mcx$ where $m$ and $c$ have the usual meanings with regard to the skew lines in simple form. (10)

(b) Find the shortest distance between the lines $\frac{x-5}{3} = \frac{y-6}{-4} = -9$;

$2x - 2y + z - 3 = 0 = 2x - y + 2x - 9$. (10)

6. (a) Find the equation of the sphere that passes through the centre of the sphere $x^2 + y^2 + z^2 + 4x - 6y + 2z - 1 = 0$ and the circle $x^2 + y^2 + z^2 + 2x - 4z - 11 = 0$,
$x + 2y + 2z - 3 = 0$. (10)

(b) Find the equation of the sphere through the circle $S = x^2 + y^2 + z^2 - 4 = 0$ and

$S_1 = x^2 + y^2 + z^2 + 4x - 2y + 4z - 10 = 0$ and

through the point $(2, 1, 1)$. (10)

7. (a) Define differentiation of a vector function, and if $r = a\cos\omega t + b\sin\omega t$ where $a, b$ are constant vectors and $\omega$ is a constant prove that $r \times \frac{dr}{dt} = \omega(a \times b)$ and $\frac{d^2r}{dt^2} + \omega^2 r = 0$. (10)

(b) Prove that grad $r^n = nr^{n-2}\bar{r}$. (10)

8. (a) Verify Gauss divergence theorem for the function $f = a(x + y)i + a(y - x)j + z^2 k$ over the hemisphere bounded by the $x \circ y$ plane and the upper half of the sphere $x^2 + y^2 + z^2 = a^2$. (10)

(b) Verify Stoke's theorem for $\overline{F} = (x^2 - y^2)\overline{i} + 2xy\overline{j}$ in the rectangular region in the $X \circ Y$ plane bounded by the lines $x = 0$, $x = a$, $y = 0$ and $y = b$. (10)

**DE–217** **23**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

MECHANICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) The resultant of two forces *P, Q* acting at a certain angle is *X* and that of *P, R* acting at the same angle is also *X*. The resultant of *Q, R* again acting at the same angle is *Y*. Prove that

$$P = \left(X^2 + QR\right)^{\frac{1}{2}} = \frac{QR(Q+R)}{Q^2 + R^2 - Y^2} \text{ and if } P + Q + R = 0,$$

then $Y = X$. (10)

(b) Explain Lami's Theorem for three forces in details. (10)

2. (a) Forces of 3, 2, 4, 5kg wt. act respectively along the sides *AB, BC, CD* and *DA* of a square. Find the magnitude of their resultant and the points where its line of action meets *AB* and *AD*. (10)

(b) Forces *P1, P2, P3* act along the sides *BC, CA, AB* of a triangle *ABC* and forces *R1, R2, R3* act along *OA, OB, OC*, where *O* is the centre of the circumcircle of the triangle ABC. Prove that the six forces are in equilibrium. (10)

3. (a) Explain the resultant of two like parallel forces acting on a rigid body. (10)

(b) State and prove Verigon's Theorem of moments. (10)

4. (a) The altitude of a right cone is $h$ and the radius of its base is $'a'$. A string is fastened to the vertex and to a point on the circumference of the circular base and is then put over a smooth peg. The cone rests with its axis horizontal. Show that the length of the string is $\sqrt{h^2 + 4a^2}$. (10)

(b) A weight can be supported on a rough inclined plane by an force $P$ acting along the plane or by a force $Q$ acting horizontally. Show that the weight is $\frac{PQ}{\sqrt{Q^2 \sec^2(\lambda) - p^2}}$, where $\lambda$ is the angle of friction. (10)

5. (a) The range of a rifle bullet is 1000m, and $\alpha$ is the angle of projection. Show that if the bullet is fired with the same elevation from a car travelling 36km/h towards the target, the range will be increased by $\frac{1000\sqrt{\tan\alpha}}{1} m$. (10)

(b) Show that, for a given velocity of projection the maximum range down an inclined plane of inclination $\alpha$ bears to the maximum range up the inclined plane of the ratio $\frac{1+\sin\alpha}{1-\sin\alpha}$. (10)

6. (a) A shot of mass m penetrates a thickness t of a fixed Plate of mass $M$. If $M$ were free to move and the resistance Supposed to be uniform, show that the thickness Penetrated is $\frac{Mt}{M+m}$. (10)

(b) Two equal billiard balls are in contact on a smooth table and a third equal ball moving along their common tangent strikes them simultaneously. Prove that $\frac{3}{5}\left(1-e^2\right)$ of its kinetic energy is lost by the impact, e being the coefficient of restitution for each pair of balls. (10)

7. (a) Show that the resultant of two simple harmonic motions in the same direction and of equal periodic time, the amplitude of one being twice that of the other and its phase a quarter of a period in advance, is a simple harmonic motion of amplitude $\sqrt{5}$ times that of the first and whose phase is in advance of the first by $\frac{\tan^{-1} 2}{2\pi}$ of a period. (10)

(b) A string of natural length a, is stretched on a smooth horizontal, table between two fixed points at a distance *na* apart; a particle of mass m is attached to the middle point of the string. If the particle is displace towards one of the fixed Points through a distance not exceeding $\frac{n-1}{2}$ a and then let go, show that it will execute Simple Harmonic Motion(SHM) whose period is independent of *n*. (10)

8. (a) Derive the period of oscillation of a *simple pendulum*. (10)

(b) Derive the *pedal equation* of a central orbit. (10)

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**DE–218** **24**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

(5 × 20 = 100)

1. (a) On any non-empty set $M$ we define $d$ as follows $d(x,y)=\begin{cases}0 & \text{if } x=y\\ 1 & \text{if } x\neq y\end{cases}$. Then prove that $d$ is a metric on $M$.

   (b) Prove that in any metric – space the intersection of a finite number of open sets is open.

   (c) Let $(m,d)$ be a metric space. Show that every subset of $M$ is open iff $\{x\}$ is open for all $x\in M$.

   (8 + 6 + 6 = 20)

2. (a) Prove that in any metric space every closed ball is a closed set.

   (b) Let $M$ be a metric space and $A\subseteq M$. Then prove that $\overline{A}=A\cup D(A)$. (10 + 10 = 20)

3. (a) Let $M$ be a metric space. Let $A$ be a connected subset of $M$. If $B$ is a subset of $M$ such that $A\subset B\subset\overline{A}$ then prove that $B$ is connected. In particular prove that $\overline{A}$ is connected.

   (b) Prove that any complete metric space is of second category. (10 + 10 = 20)

4. (a) Let $A, B$ be subsets of $R$. Prove that $\overline{A \times B} = \overline{A} \times \overline{B}$.

(b) Prove that any totally bounded metric space is separable.

(c) Prove that a non empty subset of a totally bounded set is totally bounded. (10 + 6 + 4 = 20)

5. (a) Prove that any compact subset $A$ of a metric space $(M, d)$ is closed.

(b) Prove that any closed interval $[a, b]$ is a compact subset of $R$.

(c) If a metric space $M$ is compact then prove that any infinite subset of $M$ has a limit point. (8 + 8 + 4 = 20)

6. (a) Prove that a continuous image of a compact metric space is compact.

(b) Prove that any continuous mapping $f$ defined on a compact metric space $(M_1 d_1)$ into any other metric space $(M_2 d_2)$ is uniformly continuous on $M_1$. (10 + 10 = 20)

7. (a) State and prove Weierstrass M-test.

(b) Test for uniform convergence of the series $\sum_{n=1}^{\infty} \left( \frac{1}{1+n^2 x} \right)$ on $(1, \infty)$.

(c) State and prove Cauchy criterion for uniform convergence. (6 + 6 + 8 = 20)

8. (a) $T : [0, 1/3] \to [0, 1/3]$ defined by $T(x) = x^2$. Prove that $T$ is a contraction mapping.

(b) State and prove contraction mapping theorem. (8 + 12 = 20)

**DE–219** **25**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

PROBABILITY AND STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Let S be a sample space. Let $A \subseteq S$. Then prove that

   (i) $P(A) = 1 - P(\overline{A})$, where $\overline{A}$ is the complement of A.

   (ii) $P(\phi) = 0$. (10)

   (b) One factory $F_1$ produces 1000 articles, 20 of them being defective, second factory $F_2$ produces 4000 articles, 40 of them being detective and third factory $F_3$ produces 5000 articles, 50 of them being detective. All these articles are put in one stock pile. One of them is chosen and is found to be defective. What is the probability that it is from the factory $F_1$. (10)

2. (a) A random variable X has the following probability distribution :

| | | | | | |
|---|---|---|---|---|---|
| $X$ | = | –1 | 0 | 1 | 2 |
| $p(x)$ | = | 1/3 | 1/6 | 1/6 | 1/3 |

   Find the expectation of X. (10)

(b) Calculate the Karl Pearson's coefficient of skewness. (10)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Size : | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
| Frequency : | 3 | 6 | 9 | 13 | 8 | 5 | 4 |

3. (a) Fit a curve of the form $y = ab^x$ to the following data. (10)

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Year ($x$) : | 1951 | 1952 | 1953 | 1954 | 1955 | 1956 | 1957 |
| Production in tons ($y$) : | 201 | 263 | 314 | 395 | 427 | 504 | 612 |

(b) The coefficient of rank correlation of marks obtained by 10 students in Mathematics and Physics was found to be 0.8. It was later discovered that the differences in ranks in two subjects obtained by one of the students was wrongly taken as 5 instead of 8. Find the correct coefficient of rank correlation. (10)

4. (a) The two random variables $X$ and $Y$ have the regression lines $3x + 2y - 26 = 0$ and $6x + y - 31 = 0$. Find

(i) the mean values of $X$ and $Y$

(ii) the correlation coefficient between $X$ and $Y$.

(iii) the variance of $Y$ if the variance of X is 25. (10)

(b) Drive the formula for correlation coefficient between two random variable $X$ and $Y$. (10)

5. (a) Six dice are thrown 729 times. How many times do you expect at least 3 dice to show a five or six. (10)

(b) Prove recurrence relation for central moments for Poisson distribution is $\mu_{r+1} = r\lambda\mu_{r-1} + \lambda \dfrac{d\mu_r}{d\lambda}$. (10)

6. (a) Prove that the Mean deviation about mean for Normal distribution is $\sigma\sqrt{2/\pi}$ (approximately $(4/5)\sigma$). (10)

(b) If $X$ is normally distributed with mean 8 and S.D 4 find

(i) $P(5 \le X \le 10)$

(ii) $P(10 \le X \le 15)$

(iii) $P(X \ge 15)$

(iv) $P(X \le 15)$. (10)

7. (a) A group of 10 rats fed on a diet A and another group of 8 rats fed on a different diet B recorded the following increase in weight in gms.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Diet A : | 5 | 6 | 8 | 1 | 12 | 4 | 3 | 9 | 6 | 10 |
| Diet B : | 2 | 3 | 6 | 8 | 1 | 10 | 2 | 8 | – | – |

Test whether a significant difference between diets $A$ and $B$. (10)

(b) In one sample of 8 observations the sum of the squares of deviations of the sample values from the sample mean was 84.4 and in another sample of 10 observations it was 102.6. Test whether the difference in variances is significant at 5% level Using F-test. (10)

8. (a) A company manufacturing a certain product appointed 4 salesman $S_1$, $S_2$, $S_3$, $S_4$. In three different areas A, B and C. The quantity of the product sold is given below. Carry out the analysis of variance $(\wedge\vee)$ to test whether there is significant difference in the sales of4 sales men. (10)

| | $S_1$ | $S_2$ | $S_3$ | $S_4$ |
|---|---|---|---|---|
| A | 1600 | 1610 | 1650 | 1680 |
| B | 1580 | 1640 | 1640 | 1700 |
| C | 1460 | 1550 | 1600 | 1620 |

(b) Construct Index numbers of prices by applying Fisher's index method and Laspeyre's index method. (10)

| Commodities | Base Year | | Current Year | |
|---|---|---|---|---|
| | Price | Qty | Price | Qty |
| A | 2 | 8 | 4 | 6 |
| B | 5 | 10 | 6 | 5 |
| C | 4 | 14 | 5 | 10 |
| D | 2 | 19 | 2 | 13 |

**DE–220** **31**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) For any four sets $A, B, C$ and $D$ Prove :

   (i) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

   (ii) $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$. (10)

   (b) (i) If $\rho$ and $\sigma$ are equivalence relations defined on a set $S$, Prove that $\rho \wedge \sigma$ is also an equivalence relation. (5)

   (ii) Let $f : A \to B, g : B \to C$ be bijections, prove that $g \circ f : A \to C$ is also bijection. (5)

2. (a) Prove : If $A, B$ and $C$ are any three sets then

   (i) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

   (ii) $(A - B) \times C = (A \times C) - (B \times C)$. (10)

   (b) Prove: If A, B and C are any three sets, then

   (i) $A \Delta B = (A \cup B) - (A \cap B)$

   (ii) $A \Delta (B \Delta C) = (A \Delta B) \Delta C$. (10)

3. (a) Prove : Let A and B be two subgroups of a group G, then AB is a subgroup of G iff AB = BA. (10)

(b) Prove that a subgroup of cyclic group is cyclic. (10)

4. (a) State and prove 'fundamental theorem' of homomorphism. (10)

(b) Prove that the 'isomorphism' is an equivalence relation among groups. (10)

5. (a) Prove that the set F of all real numbers of the form $a + b\sqrt{2}$, where a, b $\in Q$ is a field under the usual addition and multiplication of real numbers. (10)

(b) Prove: Let R be an Euclidean domain, then any two elements a, b $\in$ R have a g.c.d. and it is of the form $ax + by$, where $x, y \in$ R. (10)

6. (a) State and prove division algorithm. (10)

(b) Prove that the set of all matrices of the form $\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix}$, where a $\in$ R is a ring under matrix addition and multiplication. (10)

7. (a) Prove : Let V be a vector space over a field F and S be a non-empty subset of V, then

(i) L(S) is a subspace of V

(ii) S $\subseteq$ L(S)

(iii) If W is any subspace of V such that S $\subseteq$ W then L (S) $\subseteq$ W. (10)

(b) Prove: Let V be a vector space over F and W a subspace of V. Define $V / W = \{W + v ; v \in V\}$.

Then $V / W$ is a vector space over F under the following operations :

(i) $(W + v_1) + (W + v_2) = (W + v_1 + v_2)$

(ii) $\alpha\ (W + v_1) = W + \alpha v_1$. (10)

8. (a) Prove: Let V and W be vector spaces over a field F and $T : V \to W$ be an epimorphism, then

(i) Ker (T) = $V_1$ is a subspace of V and

(ii) $V / V_1 = W$. (10)

(b) Prove: Let V be a finite dimensional vector space over a field F, A and B be subspace of V, then

dim (A + B) = dim (A) + dim (A $\cap$ B) (10)

**DE–221** **32**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) A firm manufactures two types of products *A* and *B* and sells them at a profit of Rs. 2 on type *A* and Rs. 3 on type *B*. Each product is processed on two machines M1 and M2. Type *A* requires 1 minute of processing time on M1 and 2 minutes on M2. Type B requires 1 minute on M1 and 1 minute on M2. Machine M1 is available for not more than 6 hours 40 minutes while machine M2 is available for 10 hours during any working day. Formulate the problem as a LPP so as to maximize the profit.

(b) Solve graphically the following LPP:

Minimize $Z = 20x_1 + 40x_2$

Subject to the constraints :

$$36x_1 + 6x_2 \geq 108$$
$$3x_1 + 12x_2 \geq 36$$
$$20x_1 + 10x_2 \geq 100$$
$$\text{and } x_1, x_2 \geq 0.$$

2. (a) Prove that the dual of dual LPP problem is primal.

(b) Use duality to solve the following L.P.P.

Maximise $z = 2x_1 + x_2$

Subject to the constraints :

$$x_1 + 2x_2 \le 10$$
$$x_1 + x_2 \le 6$$
$$x_1 - x_2 \le 2$$
$$x_1 - 2x_2 \le 1$$
$$x_1, x_2, x_3 \ge 0.$$

3. (a) Use penalty method to,

Minimize $z = 2x_1 + x_2$

Subject to the constraints :

$$3x_1 + x_2 = 3$$
$$4x_1 + 3x_2 \ge 6$$
$$x_1 + 2x_2 \le 3$$
$$x_1, x_2 \ge 0.$$

(b) Find the optimum integer solution to the following L.P.P.

Maximize $z = x_1 + 4x_2$

Subject to the constraints :

$$2x_1 + 4x_2 \le 7$$
$$5x_1 + 3x_2 \le 15$$
$$x_1, x_2 \ge 0 \text{ and are integers.}$$

4. (a) Use Charnes penalty method to solve the LPP:

Maximize $Z = x_1 + 2x_2 + 3x_3 - x_4$

Subject to :

$$x_1 + 2x_2 + 3x_3 = 15$$
$$2x_1 + x_2 + 5x_3 = 20$$
$$x_1 + 2x_2 + x_3 + x_4 = 10$$
$$x_1, x_2, x_3, x_4 \geq 0.$$

(b) State the cutting plane algorithm for solving integer Programming problem.

5. (a) Explain the matrix minima method to find an initial basic feasible Solution to L.P.P.

(b) Find the initial basic feasible solution for the following transportation problem by Vogel's approximation method.

| | | | |
|---|---|---|---|
| 1 | 2 | 6 | 7 |
| 0 | 4 | 2 | 12 |
| 3 | 1 | 5 | 11 |
| 10 | 10 | 10 | |

6. (a) Solve the following transportation problem.

| | To | | | | | | |
|---|---|---|---|---|---|---|---|
| From | 9 | 12 | 9 | 6 | 9 | 10 | 5 |
| | 7 | 3 | 7 | 7 | 5 | 5 | 6 |
| | 6 | 5 | 9 | 11 | 3 | 11 | 2 |
| | 6 | 8 | 11 | 2 | 2 | 10 | 9 |
| | 4 | 4 | 6 | 2 | 4 | 2 | 22 |

(b) Discuss the Hungarian method for solving an assignment problem.

7. (a) Define zero-sum game using game theory principles.

(b) Solve the following $2 \times 4$ game graphically.

Player B

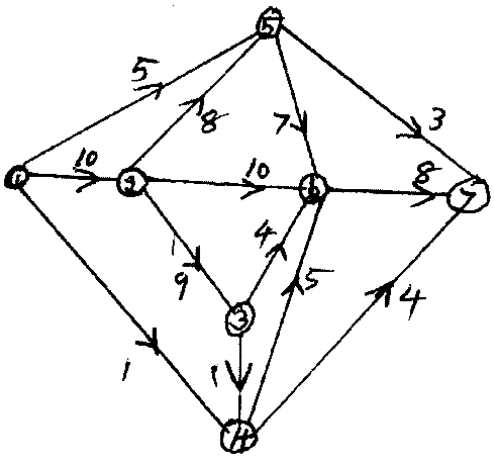
| Player A | $B_1$ | $B_2$ | $B_3$ | $B_4$ |
|---|---|---|---|---|
| $A_1$ | 2 | 1 | 0 | –2 |
| $A_2$ | 1 | 0 | 3 | 2 |

8. (a) Explain the following terms

(i) Optimistic time estimate

(ii) Pessimistic time estimate.

(b) Find the critical path for the project illustrated in the network below:



**DE–222** **33**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

(5 × 20 = 100)

1. (a) Explain iteration method. (10)

   (b) Find the real root of $x^3 - 3x + 1 = 0$ lying between 1 and 2 upto three decimal places using Newton–Raphson method. (10)

2. Solve the following system of equations using

   (a) Gauss elimination method and (10)

   (b) Gauss–Jordon method. (10)

   $x - y + z = 1$;

   $-3x + 2y - 3z = -6$

   $2x - 5y + 4z = 5$.

3. (a) Using Newton's forward interpolation formula find the value of Y when X = 7, from the following data: (10)

| | | | | |
|---|---|---|---|---|
| X: | 4 | 6 | 8 | 10 |
| Y: | 1 | 3 | 8 | 16 |

   (b) State and prove Newton's backward interpolation formula. (10)

4. (a) Explain Stirling's formula. (10)

(b) Using Everet's formula find $y$ (25) from the following data: (10)

| | | | | |
|---|---|---|---|---|
| X: | 20 | 24 | 28 | 32 |
| Y: | 2854 | 3162 | 3544 | 3992 |

5. (a) State and prove the fundamental theorem for finite difference. (10)

(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Trapezoidal rule with $h=0.2$. Hence determine the value of $\pi$. (10)

6. (a) Derive Simpson's $\frac{3}{8}$ rule for integration. (10)

(b) Evaluate $\int_0^{1/2} \frac{dx}{\sqrt{1-x^2}}$ by Weddle's rule taking $h=0.1$. Compare with the actual value and get the numerical difference between them. (10)

7. Using Taylor's series method, obtain the values of $y$ at $x=0.2$ correct to four decimal places, if $y$ satisfies the equation $\frac{d^2y}{dx^2}=xy$ given that $\frac{dy}{dx}=1$ and $y=1$ when $x=0$. (20)

8. Given $\frac{dy}{dx}=\frac{1}{2}\left(1+x^2\right)y^2$ and $y(0)=1, y(0.1)=1.06, y(0.2)=1.12, y(0.3)=1.12$. Evaluate $y(0.4)$ by Milne's predictor corrector method. (20)

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**DE–223** **34**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2014.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Prove that analytic function in a region with constant modulus is constant. (10)

   (b) Show that $f(z) = \sqrt{r}(\cos\theta/2 + i\sin\theta/2)$ where $r > 0$ and $0 < \theta < 2\pi$ is differentiable and find $f'(z)$. (10)

2. (a) Show that $u = \log \sqrt{x^2 + y^2}$ is harmonic and determine its conjugate and hence find the corresponding analytic function $f(z)$. (10)

   (b) Show that if $u$ and $v$ are conjugate harmonic functions the product $uv$ is also a harmonic function. (10)

3. (a) Prove Let $f$ be an analytic function defined in a region $D$. Let $z_0 \in D$. If $f'(z_0) \neq 0$ then $f$ is conformal map at $z_0$. (10)

   (b) Find the image of the strip $2 < x < 3$ under the map $W = \frac{1}{z}$. (10)

4. (a) Determine the bilinear transformation which maps $0, 1, \infty$ into i, $-1$, $-1$ respectively. Under this transformation show that the interior of the unit circle of the z – plane maps into the half plane left to the v axis. (10)

(b) Prove that any bilinear transformation can be expressed as a product of translation, rotation, magnification or contraction and inversion. (10)

5. State and prove Cauchy's theorem on rectangle. (20)

6. (a) Show that $f$ is analytic within and on a simple closed curve $C$ and if $z_0$ is not on C then $\int_C \frac{f'(z)(dz)}{z - z_0} = \int_C \frac{f(z)(dz)}{(z - z_0)^2}$. (10)

(b) Expand $ze^{2e}$ as a Taylor's series about $z = -1$ and determine the region of convergence. (10)

7. (a) Expand $\frac{1}{z^2 - 3z + 2}$ in Laurent's series valid in the region $1 < |z| < 2$. (10)

(b) State and prove Rouche's theorem. (10)

8. (a) Prove :

(i) Cauchy's integral formula. (5)

(ii) Residue theorem. (5)

(b) Evaluate $\int_C \frac{z+1}{z^2 + 2z + 4}$, where C is the circle $|z + 1 + i| = 2$. (10)

**DE–224** **35**

DISTANCE EDUCATION

B.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2014.

DISCRETE MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. (a) Explain : Negation, Conjunction, Disjunction, Conditional Statement and Bi-conditional statement. (10)

   (b) Construct a truth table for the following statements

   (i) $(P \to Q) \wedge (Q \to P)$

   (ii) $\neg(P \wedge Q) \rightleftarrows \neg P \vee \neg Q$

   (iii) $(Q \wedge (P \to Q)) \to P$. (10)

2. (a) Explain equivalence formula with an example. (10)

   (b) Indicates which ones are tautology or contradiction

   (i) $(P \vee \neg P) \vee Q \Leftrightarrow Q$

   (ii) $(P \vee \neg P) \Leftrightarrow Q \vee \neg Q$

   (iii) $(P \to Q) \Leftrightarrow (\neg P \vee Q)$. (10)

3. (a) Show that $R \to S$ can be derived from the premises $P \to (Q \to S)$, $\neg R \vee P$ and $Q$ . (10)

(b) Test the validity of the following argument : (10)

If the study, then I will not fail in mathematics. If I do not play basket ball, then I will study. But I failed in Mathematics. Therefore I must have played basket ball.

4. (a) Show that :

$$((\forall x)\, H(x) \to M(x)) \wedge H(S) \Rightarrow M(S).$$ (10)

(b) Show that :

$$(\exists x)(P(x) \wedge Q(x)) \Rightarrow (\exists x) P(x) \wedge (\exists x)\, Q(x).$$ (10)

5. (a) Explain incidence Matrix and Adjacency matrix with examples. (10)

(b) (i) Define directed graph and undirected graph with an example. (5)

(ii) Define a cycle graph with an example. (5)

6. (a) (i) Define spanning tree with an example. (5)

(ii) Prove that if $G$ is a tree with n vertices, the $G$ has n – 1 edges, then $G$ is a tree. (5)

(b) Prove that $K \le K' \le S$ for a graph $G$. (10)

7. (a) Prove that a graph is bipartite if and only if it contains no odd cycle. (10)

(b) Prove that if $n \ne 1$ is a positive odd integer, then the complete graph $K_n$ contains $\frac{(n-1)}{2!}$ edge – disjoint Hamilton cycles. (10)

8. (a) Explain max-flow min-cut-theorem. (10)

(b) Explain Ford and Fulkerson algorithm. (10)

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